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Image segmentation using a unified Markov random field model

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Abstract: Markov random field model (MRF) has attracted great attention in the field of image segmentation. Its basic unit can be pixels or regions. These pixel-based or region-based MRF models have their own advantages and disadvantages. In order to complement advantages of each other, a unified Markov random field (UMRF) model is proposed in this paper. The strength of the UMRF model lies in two aspects. First, the proposed model combines the benefits of the pixel-based and the region-based MRF models by decomposing the likelihood function into the product of the pixel likelihood function and the regional likelihood function. It can make the UMRF model take both pixel information and regional information into account. Second, a new regional feature is designed for the UMRF model to describe macro texture patterns. A principled probabilistic inference is developed to integrate different types of likelihood information and the spatial constraint by iteratively updating the posterior probability of the proposed model. Segmentation results can be achieved when iteration converges. Texture, remote sensing and nature images are employed to test the effectiveness of the proposed model. Experimental results illustrate that our model can achieve higher segmentation accuracy than either the pixel-based or region-based MRF models.

1 Introduction

Image segmentation is one of the most important tasks in computer vision and it has many applications in many other fields including pattern recognition, remote sensing, medical diagnostic etc. [1–5]. The purpose of segmentation is to partition an image into several homogeneous regions. Many methods have been proposed to solve this problem. For instance, the feature-based methods such as clustering [6–8] and threshold [9–11] are efficient and robust; the edge-based methods such as watershed [12–14] always have a nice performance in some certain local results; the model-based methods, such as active contour [15–17] and random field [18–21], possess the favourable mathematical foundation etc. Many works has been studied in the past, but the image segmentation is still challenging. One of its main difficulties is that a good segmentation method needs not only the effective features extracted from the image but also the additional prior knowledge, such as the contour smoothness, the contextual information, the object granularity etc.

Markov random field (MRF) model is a probabilistic graphical model, which provides a statistical way to model the spatial contextual constraints as the prior information. It has been widely employed for its intrinsic capability of resisting noise and reducing the impact of intra-class variations [19, 22–24]. This model was initially used to segment the image by defining it over pixels [18, 25]. Fig. 1 shows the structure of the pixel-based MRF model, where a circle denotes one pixel class label X_s and a square denotes the observed feature y_s of this pixel. Since spatial neighbouring relationship among pixels is regular, the pixel-based MRF model could conveniently use the spatial contextual information on the lattice. However, the calculation of the pixel-based MRF model is time-consuming, and its small neighbourhood also limits the segmentation accuracy. Hence, many researchers have tried to extend the MRF model from the pixel level to the region level [22, 26–29].

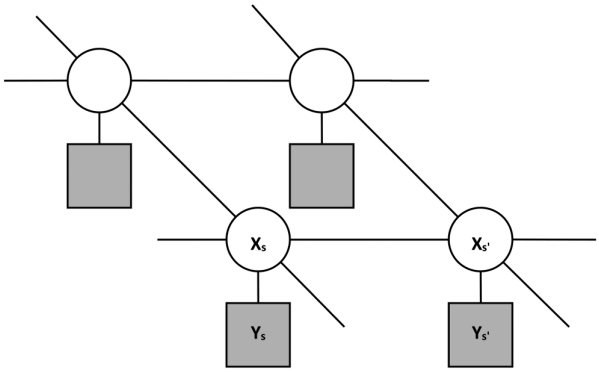


Fig. 1 *Part of a pixel-based MRF, with labels (circles) and observations (squares)*

By employing the idea of geographic object-based image analysis [30–32], the observed image is first divided into some initially over-segmented regions, and then the region-based MRF model is defined over the region adjacency graph (RAG) [21] consisted of these regions. Since each over-segmented region is used as the basic site unit in the probabilistic graph of the region-based MRF model, it can model more macro and complex spatial patterns in a large neighbourhood and reduce the calculating time. An example is illustrated in Fig. 2, where the left part is an over segmented image and the right is a part of the RAG for the left. For each region S, the circle in the right part denotes its class label X_s and the square in the left denotes corresponding regional feature y_s, such as the regional mean value that averages the spectral value of each pixel in region S. In the region-based MRF model, the initial segmentation is very important since it could directly affect the final segmentation accuracy. Different over-segmented methods, such as watershed [12–14], mean shift (MS) [33], normalised cut (Ncut) [34], and turbopixel [35], were studied to provide a better over-segmented result, and many methods were also employed to the region-based MRF model. For instance, Yu and David [27] used watershed to provide initial regions and defined the MRF over the RAG. Then the results were obtained using a region growing technique based on the edge penalty function. Xia [26] also employed the watershed to get initial regions, and used the probabilistic inference to obtain the result. Zhang and Ji [28] even incorporated region-based random field model and the Bayesian

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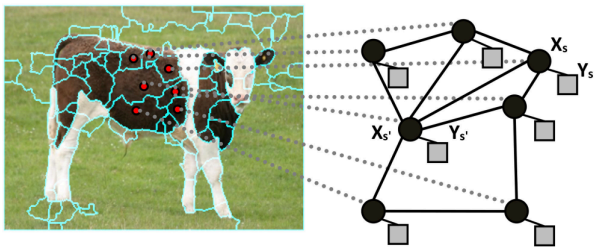


Fig. 2 *Example of a region-based MRF. Left: an over-segmented image. Right: a part of RAG with label (circles) and observation (squares)*

Network as a unified graphical model to realise image segmentation. Olding *et al.* [36] proposed the region overlap graph by combining the information of several overlapped over segmentations for the region-based MRF model. Although region-based MRF models can overcome some shortcomings of pixel-based MRF models, it still faces many challenges, such as the inaccuracy of the initial regions and the irregular spatial contextual relationship of initial regions.

For the pixel-based MRF model, it has the regular spatial context, but it is time consuming and cannot describe the long range interactions among pixels, i.e. macro texture pattern. On the contrary, the region-based MRF can capture the macro texture pattern by utilising the region-based information, but it suffers the irregular spatial context. Therefore, if one model can possess the regular spatial context and the description of the macro texture pattern at the same time, this model can take advantages of both MRF models and solve pitfalls of each model. Namely, the description of the macro texture pattern could make the pixel-based MRF model consider interactions in a large neighbourhood; the regular spatial context could make the region-based MRF model capture the spatial relationship between regions more easily. Motivated by this idea, we propose a unified MRF (UMRF) model to integrate the pixel- and the region-based MRF models. The UMRF is characterised by two aspects. First, the UMRF model extends the likelihood function to the product of the pixel and the regional likelihood functions for capturing both pixel-based and regional information. To model different information at the same probabilistic graph and use the regular spatial context, unified likelihood information and the label field are further defined on the regular lattice grid. Second, a new regional feature is introduced to capture the macro interaction between regions for the regional likelihood function of the UMRF model.

This paper is organised as follows. A brief review of the MRF model is discussed in Section 2. The new regional feature of the UMRF model is introduced in Section 3. The details of the UMRF model are given in Section 4. The algorithm is described in Section 5. The experiments are presented in Section 6 and the conclusion is given in Section 7.

2 MRF model and image segmentation

Let $\boldsymbol{S}$ denote a set of sites. $\boldsymbol{Y} = \{y_s | s \in S\}$ is the observed image defined on $\boldsymbol{S}$. $\boldsymbol{X} = \{X_s | s \in S\}$ is the label random field defined on $\boldsymbol{S}$. Each random variable X_s in $\boldsymbol{X}$ represents the class of site s and takes value from the set $\Lambda = \{1, 2, \ldots, n\}$, where n is the number of classes. Let $\boldsymbol{x} = \{x_s | s \in S\}$ denote a realisation of $\boldsymbol{X}$. In the MRF model, the image segmentation is converted into the estimation of a best realisation $\hat{\boldsymbol{x}}$ given the observed image $\boldsymbol{Y}$, i.e. $\hat{\boldsymbol{x}} = \arg\max_x P(x|\boldsymbol{Y})$.

According to the Bayesian rule, the posterior probability $P(\boldsymbol{X}|\boldsymbol{Y})$ is equal to $((P(\boldsymbol{Y}|\boldsymbol{X})P(\boldsymbol{X}))/P(\boldsymbol{Y}))$. Since the probability $P(\boldsymbol{Y})$ is a constant, the estimation of the best realisation $\hat{\boldsymbol{x}}$ can be obtained by maximising $P(\boldsymbol{Y}|\boldsymbol{X})P(\boldsymbol{X})$, which needs to determine the forms of $P(\boldsymbol{X})$ and $P(\boldsymbol{Y}|\boldsymbol{X})$.

First, the joint probability $P(\boldsymbol{X})$ is used to model the label random field $\boldsymbol{X}$. Since $\boldsymbol{X}$ is assumed to possess the Markov property in the MRF model, $P(\boldsymbol{X})$ is of Gibbs distribution according to the theorem of Hammersley–Clifford [21]. That is $P(\boldsymbol{X} = \boldsymbol{x}) = (1/Z)\exp(-U(\boldsymbol{x}))$, where $Z = \sum_{\boldsymbol{x}} U(\boldsymbol{x})$ is the normalisation factor and $U(\boldsymbol{x}) = \sum_{s \in S} U(x_s, x_{N_s})$ is the energy function. Here $U(x_s, x_{N_s}) = \sum_{t \in N_s} V(x_s, x_t)$, where N_s is set of sites neighbouring site s and each $V(x_s, x_t)$ is the potential function between site s and site t, $t \in N_s$. The multilevel logistic (MLL) model [21] is usually employed to define the potential function $V(x_s, x_t)$, which is

$$V(x_s, x_t) = \begin{cases} \beta & \text{if } x_s \neq x_t \\ -\beta & \text{if } x_s \neq x_t \end{cases}, \quad (1)$$

where $\beta > 0$ is the potential parameter and $t \in N_s$. Based on the MLL model, $P(\boldsymbol{X})$ would have a large value if the local neighbour labels are same, otherwise small. This characteristic encourages the adjacent pixels to be classified into the same label, which would make the MRF model resist noise and reduce the impact of intra-class variations.

Second, the likelihood function $P(\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x})$ is a conditional probability function. It is used to measure the probability that how the observed image $\boldsymbol{Y}$ match a given realisation $\boldsymbol{X} = \boldsymbol{x}$. The sites of the likelihood function are assumed to be independent when the label field is given. That is

$$P(\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x}) = \prod_{s \in S} P(y_s|x) = \prod_{s \in S} P(y_s|x_s).$$

For the pixel-based MRF model, each site $s = (i, j) \in \boldsymbol{S}$ in the above equation denotes a pixel, and its feature y_s is the vector that consists of spectral values of each band at site s, i.e. $y_s = (y_s^1, y_s^2, \ldots, y_s^D)$. Here, y_s^i denotes the value of spectral band i, and D is the number of spectral bands for the observed image. The Gaussian distribution is usually employed to model $P(y_s|x_s)$ in the above equation, i.e.

$$P(y_s|x_s) = (2\pi)^{-D/2}|\Gamma|^{-1/2}\exp\left(-\frac{1}{2}(y_s - \mu)^{\mathrm{T}}\Gamma^{-1}(y_s - \mu)\right),$$

where $\mu = E(X_s)$, and Γ is the covariance matrix of X_s.

At the pixel-based MRF model, the site set $\boldsymbol{S} = \{s | 1 \leq i \leq M, 1 \leq j \leq N\}$ is an $M \times N$ discrete rectangular lattice. Due to the regular spatial context, one can conveniently define the neighbourhood system $\boldsymbol{N} = \{N_s | s \in S\}$ as the four- or the eight-neighbourhood system. However, the pixel-based neighbourhood relationship is limited to describe the macro texture pattern of an image. To model complex and macro texture patterns, the multi-resolution MRF (MRMRF) model is introduced into the MRF model [19, 21, 24, 37]. The MRMRF model improves the MRF model by considering the image information at different resolutions. Although the MRMRF model can describe interactions in a lager neighbourhood compared with the MRF model, it is still a pixel-based method. This would lead to some misclassifications and limit the segmentation accuracy.

The region-based MRF models are proposed to improve the MRF model by using over-segmented regions as the basic site unit. They are defined on the site set $\boldsymbol{R} = \{r\}$ of regions, where each r is an over-segmented region and $\bigcup_{r \in R} r = \boldsymbol{Y}$, $r \cap t = \emptyset (\forall r \neq t$, and $r, t \in \boldsymbol{R})$. It could make the region-based MRF model to describe more macro spatial interactions. However, the region-based MRF model also suffers disadvantages. First, the initially over segmented regions may be imprecise. As mentioned in [35], the approaches used for initially segmentation, such as watershed, have some imprecise segments that would lead to the unfavourable result. Second, the spatial context relationships between regions are irregular. It will be difficult to define the neighbourhood system for these regions.

3 Regional feature extraction

The UMRF model is studied in this paper to unify the benefits of pixel- and region-based MRF models and break their limitations. It will use both the pixel feature $\boldsymbol{Y}^{\mathrm{P}}$ and the regional feature $\boldsymbol{Y}^{\mathrm{R}}$. In

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Fig. 3 *Demonstration of the over segmented image and its corresponding regional feature.*
(a) Original image, ***(b)*** Over segmented image using MS, ***(c)*** The regional feature Y_s^{R}

the UMRF model, the pixel feature $\boldsymbol{Y}^{\mathrm{P}}$ is defined as the spectral value of pixels since the information of each pixel is limited. However, an over-segmented region, besides the spectral feature, includes more information, such as the shape, the size, the region context etc. Hence, it is critical to extract regional feature $\boldsymbol{Y}^{\mathrm{R}}$ incorporating various valuable information.

To extract the regional feature, the proposed model first obtains these initial regions using the MS, an example is shown in Fig. 3*b*. Although regional mean value and variance are the two most used features in the region-based MRF models [26, 38], we will not employ them in this paper since the spectral value has been considered for $\boldsymbol{Y}^{\mathrm{P}}$. Since the region context and region size can complement the spectral value information, they are used to design the regional feature $\boldsymbol{Y}^{\mathrm{R}}$ for the UMRF model. Namely, for each site s, its regional feature Y_s^{R} is defined as follows:

$$Y_s^{\mathrm{R}} = [p_{Rs} - p_{Rs} \cdot \log(p_{Rs})] + \frac{1}{|N_{Rs}|} \sum_{T \in N_{Rs}} [p_T - p_T \cdot \log(p_T)] \quad (2)$$

where Rs denotes the initial over-segmented region including s, p_{Rs} is the area ratio of the region Rs to the whole image, and N_{Rs} is the set of neighbour regions of Rs.

The details that how $\boldsymbol{Y}^{\mathrm{R}}$ uses the region size and the region context are discussed below. First, due to the function $f(x) = x - x \cdot \log(x)$ is monotonously increased for $0 \le x \le 1$, the term $f(p_{Rs}) = [p_{Rs} - p_{Rs} \cdot \log(p_{Rs})]$ in (2) can reflect the size of the region Rs. Second, the average value $(1/|N_{Rs}|)\sum_{T \in N_{Rs}}[p_T - p_T \cdot \log(p_T)]$ of (2) could reflect the region context. Namely, $(1/|N_{Rs}|)\sum_{T \in N_{Rs}}[p_T - p_T \cdot \log(p_T)]$ can take the influence of the neighbour region size into account. If the sizes of neighbour regions are large, they will increase Y_s^{R}. On the contrary, if the sizes of neighbour regions are small, they will decrease Y_s^{R}. Furthermore, $(1/|N_{Rs}|)\sum_{T \in N_{Rs}}[p_T - p_T \cdot \log(p_T)]$ can show the consistence of neighbour regions. For example, we assume that regions A and B are the neighbourhoods of region Rs, and regions A' and B' are neighbouring to region Rs'. Furthermore, if size(Rs) = size(Rs'), size($A + B$) = size($A' + B'$), size(A) = size(B), but size(A') ≠ size(B'), then we have $(1/2)[(p_A - p_A \cdot \log p_A) + (p_B - p_B \cdot \log p_B)] > (1/2)[(p_{A'} - p_{A'} \cdot \log p_{A'}] + (p_{B'} - p_{B'} \cdot \log p_{B'})]$ since the function $f(x) = x - x \cdot \log(x)$ is concave. That is to say, different sizes of neighbouring regions can affect the regional feature Y_s^{R}. Therefore, the proposed regional feature Y_s^{R} can describe regional contextual information effectively. An example is shown in Fig. 3*c*, where the bright part means that the Y_s^{R} has a large value and usually represents a large region, and the dark part denotes the Y_s^{R} has a small value and indicates a small region. Moreover, if a large region is surrounded by small regions, the Y_s^{R} will decrease, such as the forehead of the human; conversely, if a small region is surrounded by large regions, the Y_s^{R} will increase, such as the curtain.

4 Proposed model

4.1 Overview of the model

Both the pixel-based MRF and the region-based MRF have advantages and disadvantages as mentioned above. To overcome these shortcomings, one feasible way is to complement each other's advantages of these two types of MRF models. Based on this idea, the UMRFmodel is proposed in this section. The flow chart of the proposed model is shown in Fig. 4. The UMRF model first extracts the pixel feature $\boldsymbol{Y}^{\mathrm{P}}$ and the regional feature $\boldsymbol{Y}^{\mathrm{R}}$. Then, it combines these features together by the likelihood function and integrates them with the spatial information of the posterior probability. Finally, the segmentation result can be provided by solving the MAP estimation problem. The details of the UMRF are discussed in Section 4.2.

4.2 UMRF model

According to Section 2, the main advantage of the pixel-based MRF model is of the regular spatial context, and the main advantage of the region-based MRF model is to describe the macro texture pattern. The UMRF incorporates them together. First, the UMRF model assumes that the observed image $\boldsymbol{Y}$ is composed of the pixel feature $\boldsymbol{Y}^{\mathrm{P}} = \{Y_s^{\mathrm{P}} | s \in S\}$ and the regional feature $\boldsymbol{Y}^{\mathrm{R}} = \{Y_s^{\mathrm{R}} | s \in S\}$, i.e. $\boldsymbol{Y} = (\boldsymbol{Y}^{\mathrm{P}}, \boldsymbol{Y}^{\mathrm{R}})$. It makes the UMRF can describe both the micro and macro texture pattern of images by using the $\boldsymbol{Y}^{\mathrm{P}}$ and $\boldsymbol{Y}^{\mathrm{R}}$, respectively. Then, with the purpose of using the regular spatial context, the proposed model is further defined on the 2D lattice grid. To consider the effects of the regional feature and the pixel feature separately, we assume that they are independent when labels are given, and the best realisation $\hat{\boldsymbol{x}}$ of the UMRF model can be obtained as follows:

$$\begin{aligned} \hat{\boldsymbol{x}} &= \arg\max_{x} \; P(\boldsymbol{X}|\boldsymbol{Y}) \\ &= \arg\max_{x} \; P(\boldsymbol{Y}|\boldsymbol{X})P(\boldsymbol{X}) \\ &= \arg\max_{x} \; P(\boldsymbol{Y}^{\mathrm{P}}, \boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})P(\boldsymbol{X}) \\ &= \arg\max_{x} \; P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X})P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})P(\boldsymbol{X}) \end{aligned} \quad (3)$$

In (3), the likelihood function $P(\boldsymbol{Y}|\boldsymbol{X})$ is decomposed into the product of $P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X})$ and $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$. The pixel likelihood function $P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X})$ is used to model the micro texture pattern and the detailed information based on the pixel feature, and the region likelihood function $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$ is designed to model the macro texture pattern according to the regional feature. The independence assumption between them simplifies the calculation; however, the pixel feature and the regional feature should have interactions between them. Hence, we design a special parameter α in $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$ to show their relationship. Details about this parameter will be discussed in the next paragraph.

For solving the MAP estimation of (3), we first need to determine the distributions of $P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X})$, $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$, and $P(\boldsymbol{X})$. In this paper, we assume that different $Y_s^{\mathrm{P}}(s \in S)$ are independent given the labels, i.e. $P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X}) = \prod_{s \in S} P(Y_s^{\mathrm{P}}|X_s = h)$, and each $P(Y_s^{\mathrm{P}}|X_s = h)$ is of the Gaussian distribution, that is

$$P(Y_s^{\mathrm{P}}|X_s = h) = \frac{1}{(2\pi)^{D/2}\sqrt{\det(\Sigma_h^P)}} \exp\left(-\frac{1}{2}(Y_s^{\mathrm{P}} - \mu_h^P)^{\mathrm{T}} \cdot (\Sigma_h^P)^{-1} \cdot (Y_s^{\mathrm{P}} - \mu_h^P)\right). \quad (4)$$

In the above equation, the pixel feature of site s, $Y_s^{\mathrm{P}} = (y_s^1, y_s^2, \ldots, y_s^D)$, is also defined as a vector that consists of spectral values of each band, which is the same as y_s of the pixel-based MRF model in Section 2. $\theta_h^P = (\mu_h^P, \Sigma_h^P)$ is the parameter of Gaussian distribution and $\det(\Sigma_h^P)$ is the determinant of Σ_h^P, $1 \le h \le n$. To effectively integrate the pixel and region likelihood information, the UMRF model considers them at the same probabilistic graph. Hence, $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$ is defined on the pixel-based

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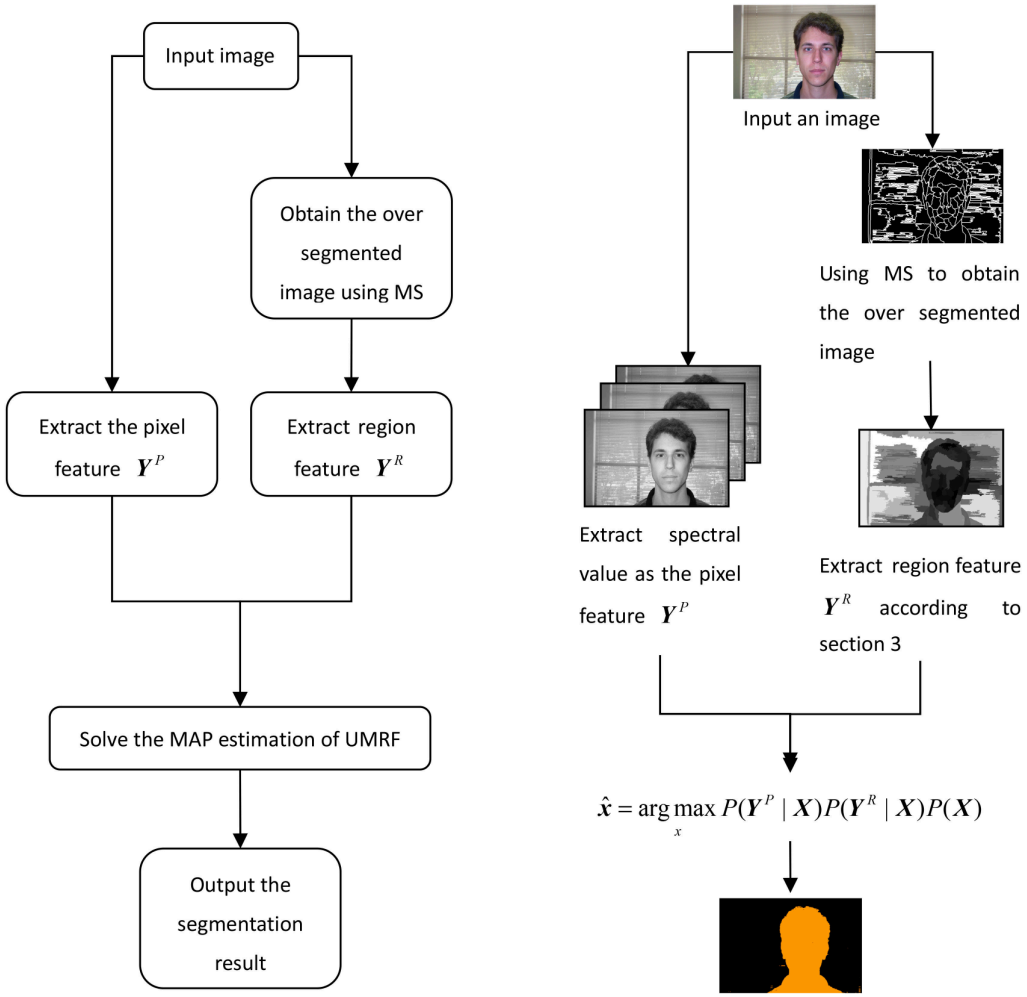


Fig. 4 *Flowchart of the proposed model (left) and a corresponding example (right)*

lattice grid as well. Different $Y_s^{\mathrm{R}}(s \in S)$ are also assumed to be independent given the labels and $P(Y_s^{\mathrm{R}} | X_s = h)$ is of Gaussian distribution with an additional parameter α, that is

$$P(Y_s^{\mathrm{R}} | X_s = h) = \frac{\sqrt{\alpha}}{(2\pi)^{D'/2}\sqrt{\det(\Sigma_h^R)}} \exp\left(-\frac{1}{2}(Y_s^{\mathrm{R}} - \mu_h^R)^T \cdot (\Sigma_h^R/\alpha)^{-1} \cdot (Y_s^{\mathrm{R}} - \mu_h^R)\right) \quad (5)$$

where D' is the dimension of Y_s^{R}, $\theta_h^R = (\mu_h^R, \Sigma_h^R)$ is the parameter of Gaussian distribution, $\det(\Sigma_h^R)$ is the determinant of Σ_h^R, $1 \le h \le n$, and the parameter α is specially designed to reflect the interaction between the pixel feature and regional feature by adjusting Σ_h^R. How α can work will be discussed in Section 4.3.

The joint probability $P(\boldsymbol{X})$ equals to $\prod_{s \in S} P(X_s | X_{N_s})$ since it possesses the Markov property. For $P(X_s | X_{N_s})$, it is defined as the MLL model mentioned at Section 2. Since the UMRF model is defined on the 2D lattice, we would define N_s as the eight-neighbourhood system for the MLL model [21]. Based on the forms of $P(\boldsymbol{Y}^{\mathrm{P}} | \boldsymbol{X})$, $P(\boldsymbol{Y}^{\mathrm{R}} | \boldsymbol{X})$, and $P(\boldsymbol{X})$, we can get the best realisation $\hat{\boldsymbol{x}} = \{\hat{x}_s | s \in S\}$ by sequentially solving each $\hat{x}_s$, i.e.

$$\hat{x}_s = \arg\max_{x_s} P(Y_s^{\mathrm{P}} | X_s = x_s) P(Y_s^{\mathrm{R}} | X_s = x_s) P(X_s = x_s | X_{N_s}), \quad (6)$$

and $\hat{x}_s$ can be iteratively optimised by maximising local conditional probabilities, which is similar as the iterated conditional mode (ICM) [25]. The algorithm of solving the $\hat{x}_s$ is given in Section 5. Note that, as mentioned above, the proposed UMRF model try to combine the benefits of the pixel- and region-based MRF models in terms of the likelihood function. However, the basic site units are different for these two types of MRF models. One is the pixel and the other is the over-segmented region obtained by the MS. To integrate the advantages of pixel and region features at the same unit, the UMRF model assigns features of an over-segmented region to each pixel in this region, i.e. Y_s^{R} defined in (2). The Bayesian framework further considers both types of features pixel by pixel, namely Y_s^{P} and Y_s^{R}; and realises segmentation by sequentially optimising each $\hat{x}_s$ in (6).

4.3 Parameters setting

There are mainly four parameters in the UMRF, i.e. $\theta_h^P = (\mu_h^P, \Sigma_h^P)$, $\theta_h^R = (\mu_h^R, \Sigma_h^R)$, α and β. The first three parameters are employed for $P(\boldsymbol{Y} | \boldsymbol{X})$, and the last potential parameter β, as mentioned in (1), is used for $P(\boldsymbol{X})$. Although some can be estimated automatically, other parameters need to be set empirically.

For $\theta_h^P = (\mu_h^P, \Sigma_h^P)$ and $\theta_h^R = (\mu_h^R, \Sigma_h^R)$, they are the mean value and variance for the Gaussian distribution, which can be automatically estimated as follows:

$$\hat{\mu}_h^P = \frac{1}{|X^h|}\sum_{s \in X^h} Y_s^{\mathrm{P}}, \quad \hat{\Sigma}_h^P = \frac{1}{|X^h|}\sum_{s \in X^h} (Y_s^{\mathrm{P}} - \mu_h^P)^{\mathrm{T}}(Y_s^{\mathrm{P}} - \mu_h^P), \quad (7)$$

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Fig. 5 *Segmentation example with different α*
(a) Original image, ***(b)*** Pixel based, ***(c)*** $\alpha = 0.25$, ***(d)*** $\alpha = 1.1$, ***(e)*** $\alpha = 4$, ***(f)*** Region based

$$\hat{\mu}_h^R = \frac{1}{|X^h|}\sum_{s\in X^h} Y_s^{\mathrm{R}}, \quad \hat{\Sigma}_h^R = \frac{1}{|X^h|}\sum_{s\in X^h}(Y_s^{\mathrm{R}} - \mu_h^R)^{\mathrm{T}}(Y_s^{\mathrm{R}} - \mu_h^R), \quad (8)$$

where X^h is the set of sites with label $X_s = h$, $1 \le h \le n$.

The values of α and β are set empirically, but there are some basic rules to guide the setting. Generally speaking, α is used to balance the influence between $P(\boldsymbol{Y}^{\mathrm{P}}|\boldsymbol{X})$ and $P(\boldsymbol{Y}^{\mathrm{R}}|\boldsymbol{X})$ for $P(\boldsymbol{Y}|\boldsymbol{X})$, β is used to balance the influence between $P(\boldsymbol{Y}|\boldsymbol{X})$ and $P(\boldsymbol{X})$. For α, it can adjust the variance of $P(Y_s^{\mathrm{R}}|X_s = h)$ for the fixed $\theta_h^R = (\mu_h^R, \Sigma_h^R)$. Large α will reduce the variance, which could stress the difference of the regional feature and lead $P(Y_s^{\mathrm{R}}|X_s = h)$ to domain the likelihood function $P(\boldsymbol{Y}|\boldsymbol{X})$. On the contrary, small α will enlarge the variance, which could tolerate the difference of the regional feature and emphasize $P(Y_s^{\mathrm{P}}|X_s = h)$ of $P(\boldsymbol{Y}|\boldsymbol{X})$. Hence, α can adjust the influence between the pixel feature and the regional feature. Fig. 5 gives an example. Fig. 5*a* is an original image. Figs. 5*b* and *f* are the segmentation results with only pixel feature and regional feature, respectively. Fig. 5*c* is the result of the UMRF with $\alpha = 0.25$. Since the value of α is small, its segmentation result is similar as the pixel-based result in Fig. 5*b*. Meanwhile, Fig. 5*e* is the result of the UMRF model with $\alpha = 4$, which is similar as the region-based result in Fig. 5*f* as this α value is large. However, choosing an appropriate $\alpha = 1.1$ based on the given image, the result of the UMRF in Fig. 5*d* will balance the impacts between the pixel- and the region-based features and provide the nice performance. For β, it is the traditional potential parameter. Similar to α, large β can lead to segmentation results with large homogeneous objects by emphasising $P(\boldsymbol{X})$. On the contrary, small β is suitable for image with many details by emphasising $P(\boldsymbol{Y}|\boldsymbol{X})$.

5 Algorithm for the proposed model

In practical implementation, this proposed model firstly gets over segmented regions using the MS provided by Edison [39], then extracts the pixel and regional feature, finally obtains the segmentation result. The algorithm is shown as follows:

Algorithm: **Input**: the observed image $\boldsymbol{Y}$ and the number of classes n
Output: the segmentation result
Steps:

1. Extract the spectral value of $\boldsymbol{Y}$ as the pixel feature $\boldsymbol{Y}^{\mathrm{P}}$;
2. Use the MS to get over segmented regions, then extract the related feature by (2) as the regional feature $\boldsymbol{Y}^{\mathrm{R}}$;
3. Get the initial label field X^0 by using the k-means algorithm to cluster the features into n classes;
4. Start iteration. Set flag = 0;
5. Use (7) and (8) to calculate the means and variances of pixel and region observations for all the n classes given X^{flag};
6. Calculate the $P(Y_s^{\mathrm{P}}|X_s = h)$, $P(Y_s^{\mathrm{R}}|X_s = h)$, and $P(X_s|X_{N_s})$ based on (1), (4), and (5);
7. Obtain the $\hat{x}_s = \arg\max_{X_s\in\{1,2,\ldots,n\}} P(Y_s^{\mathrm{P}}|X_s = h)\cdot P(Y_s^{\mathrm{R}}|X_s = h)\cdot P(X_s|X_{N_s})$ and the best realisation $\hat{\boldsymbol{x}} = \{\hat{x}_s|s\in S\}$;
8. Set $X^{\mathrm{flag}+1} = \hat{\boldsymbol{x}}$. if $X^{\mathrm{flag}+1} == X^{\mathrm{flag}}$, stop and $X^{\mathrm{flag}+1}$ is the final result; else, set flag = flag + 1, repeat the steps 5, 6, 7, and 8.

6 Experimental results

The proposed UMRF model provides a new MRF-based way to improve the segmentation accuracy by combine the pixel- and the region-based MRF models. To experimentally evaluate the proposed method, various images from different data sets are used in the experiments. Two modules are discussed in this section. First, we test the effectiveness of the OMRF-A model by comparing our model with other MRF-based or state-of-the-art methods in terms of the figure/ground segmentation. Second, we discuss the robustness of the UMRF model by testing more images from different datasets in Section 6.2.

6.1 Figure/ground image segmentation

In this section, the proposed UMRF model is compared with other state-of-the-art methods for the figure/ground segmentation. They are (i) the graph cuts (GC) method [40, 41], (ii) the MRF model using the ICM [25], (iii) the MRMRF model in wavelet domain [19, 42], (iv) the simple unsupervised MRF (SMRF) model [38], and (v) the region-based graph cut (RGC) model [43].

The GC method is a widely used approach for image segmentation by effectively solving the energy minimisation problem. The ICM is a classical pixel-based MRF model. The MRMRF is a multi-resolution model. The SMRF model is a region-based MRF defined on the RAG. The RGC model introduces the boundary regularisation with region-based properties to the segmentation energy optimised by GC. Therefore, the comparison between them and the UMRF can demonstrate the difference between our model and other MRF-based or state-of-the-art methods.

Tested images and segmentation results obtained from six different methods are demonstrated in Figs. 6 and 7. In these ten experimental groups labelling from *a* to *j*, we have performed the experiments using different types of images from the Caltech database (Groups a and b) [44], the Microsoft dataset MSRCORID (Groups c and d) and materials database (Groups g and h) [23], the Berkeley segmentation dataset (Groups e and f) [45] and the remote sensing image (Groups i and j). Tested images have various shapes and appearances, such as materials, cows, human faces and harbour. This brings challenges to the segmentation.

As we observe, the proposed method generally shows a better performance compared with other methods in Figs. 6 and 7. For instance, the appearances of human in Fig. 6(*1a*) have significant variations, hence GC, ICM, and SMRF fail to recognise the object completely. Although MRMRF could recognise the human, however, some background areas are wrongly recognised as the human face. The result of RGC also fails to recognise the hair of the human. On the contrary, the proposed UMRF model takes not only the pixel-based spectral feature but also the region size and the region contextual information into account, which leads to a better result. Furthermore, for the remote sensing image of Fig. 7(*1j*), the appearance of buildings is similar to the appearance of the bare red land, and the shadows in the urban area are quite different with the buildings. The result of GC, as shown in Fig. 7(*3j*), wrongly recognises some bare red land as the urban. Results of ICM, MRMRF, and SMRF in Figs. 7(*4j*)–(*6j*) classify shadow areas of the urban to the background. The RGC provides a result with large consistent regions in Fig. 7(*7j*) by considering the regional term, but some bare lands near the urban are wrongly classified. The result of UMRF shows a better performance than other's since it uses the micro and macro information simultaneously by taking the advantages of the pixel- and region-based MRF models.

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Fig. 6 *Figure/ground segmentations of different methods*

(1a–e) Original images, ***(2a–e)*** Artificial results, ***(3a–e)*** Results of Graph cuts, ***(4a–e)*** Results of ICM, ***(5a–e)*** Results of MRMRF, ***(6a–e)*** Results of SMRF, ***(7a–e)*** Results of region-based graph cuts, ***(8a–e)*** Results of UMRF

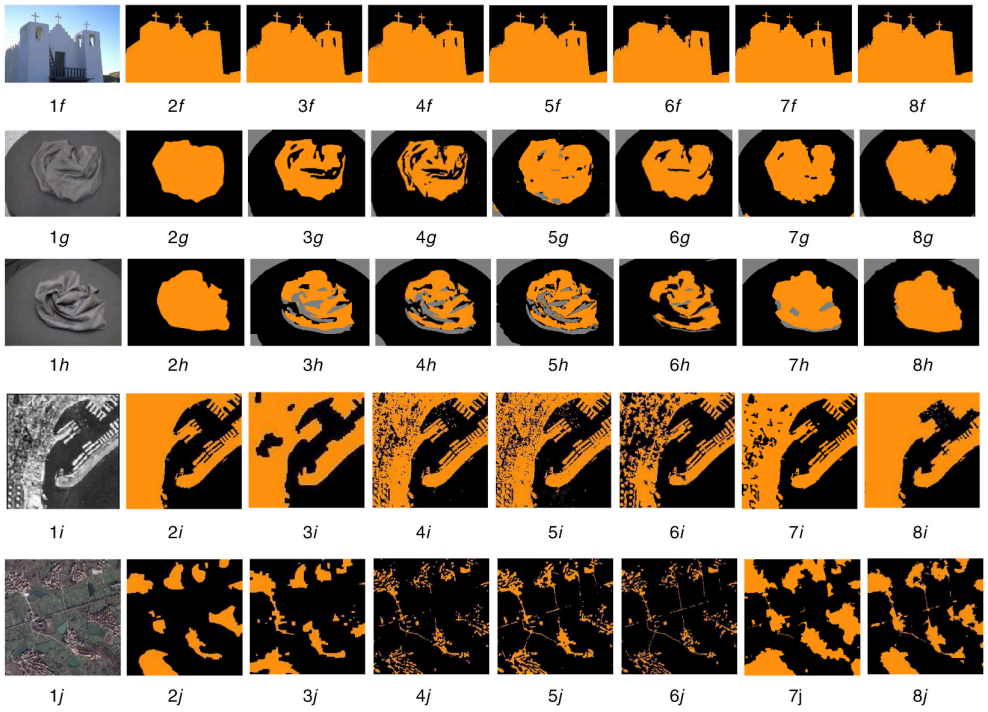


Fig. 7 *Robustness test with more figure/ground experiments*

(1f–j) More original images, ***(2f–j)*** Artificial results, ***(3f–j)*** Results of graph cuts, ***(4f–j)*** Results of ICM, ***(5f–j)*** Results of MRMRF, ***(6f–j)*** Results of SMRF, ***(7f–j)*** Results of region-based graph cuts, ***(8f–j)*** Results of UMRF

To quantitatively evaluate above experiments, the results of these six methods have been measured by the probabilistic rand index (PRI), global consistence error (GCE) [46], variation of information (VoI), and boundary displacement error (BDE) [45]. Table 1 shows the details. From the values of different quantitative indexes, they demonstrate that our model improves the segmentation accuracy. Here, large values of PRI and small values of GCE, VoI, and BDE indicate better reliability. The whole indexes of the experiments are figured in Fig. 8.

6.2 Texture and nature image segmentation

The proposed model is further tested with more images from various datasets. These tested images are shown in Fig. 9, where the texture images are from the Prague texture segmentation datagenerator (*a* and *b*) [47], and the nature images are from the Berkeley segmentation dataset (*c* to *h*) and the Microsoft dataset (MSRC2) (*i* to *n*). For some images, the within-class appearances have significant variations, such as the top-left texture in (*a*), the zebra in (*c*), the book in (*i*), and the building in (*n*). For some images, different classes have similar appearances, such as the sky

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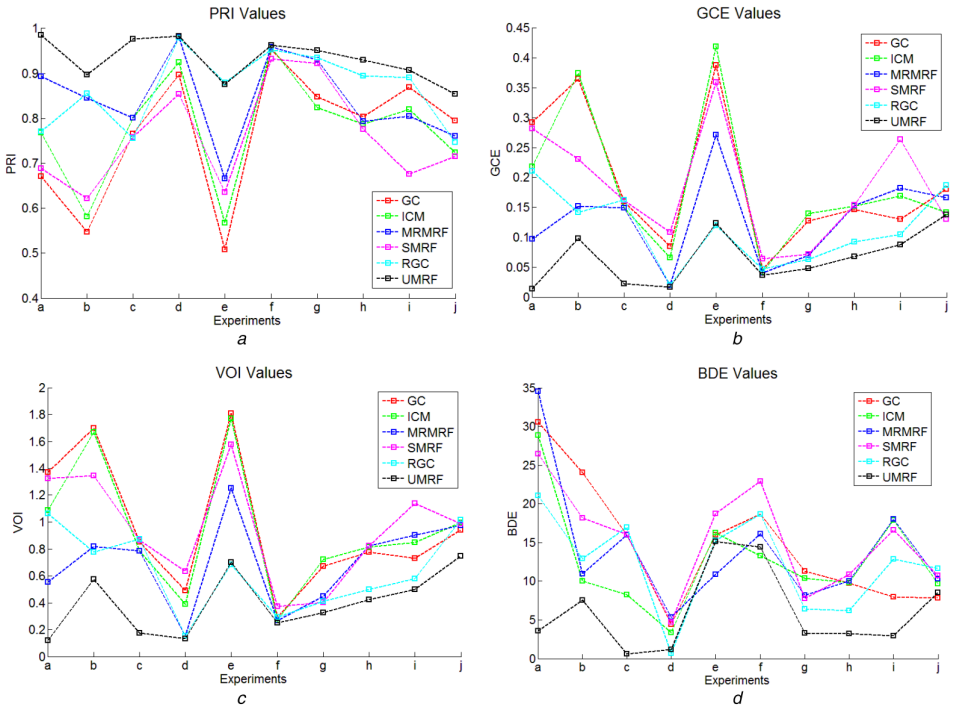


Fig. 8 *Quantitative evaluation indexes of experiments of Figs. 6 and 7*
(a) PRI, ***(b)*** GCE, ***(c)*** VoI, ***(d)*** BDE

Table 1 The values of quantitative indexes for experiments of Figs. 6 and 7

		Fig. 6				Fig. 7					
		a	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*
PRI	GC	0.671	0.547	0.766	0.897	0.508	0.952	0.848	0.804	0.869	0.795
	ICM	0.769	0.581	0.801	0.925	0.567	0.958	0.824	0.787	0.819	0.724
	MRMRF	0.893	0.845	0.801	0.981	0.666	0.959	0.930	0.793	0.805	0.761
	SMRF	0.688	0.622	0.759	0.854	0.636	0.933	0.922	0.776	0.676	0.715
	RGC	0.771	0.855	0.756	0.980	0.880	0.952	0.935	0.894	0.890	0.746
	UMRF	0.986	0.898	0.977	0.983	0.876	0.962	0.951	0.929	0.907	0.854
GCE	GC	0.292	0.365	0.160	0.085	0.388	0.047	0.127	0.146	0.130	0.181
	ICM	0.219	0.375	0.149	0.066	0.420	0.041	0.140	0.153	0.169	0.142
	MRMRF	0.097	0.152	0.150	0.019	0.272	0.040	0.070	0.153	0.183	0.166
	SMRF	0.282	0.231	0.162	0.108	0.359	0.064	0.072	0.154	0.264	0.130
	RGC	0.211	0.142	0.162	0.020	0.120	0.047	0.063	0.093	0.105	0.187
	UMRF	0.014	0.098	0.023	0.016	0.124	0.037	0.048	0.068	0.088	0.137
VOI	GC	1.371	1.698	0.851	0.492	1.809	0.295	0.670	0.776	0.730	0.941
	ICM	1.087	1.671	0.784	0.389	1.773	0.272	0.721	0.815	0.847	0.995
	MRMRF	0.554	0.819	0.787	0.154	1.253	0.267	0.447	0.821	0.904	0.976
	SMRF	1.325	1.347	0.866	0.632	1.578	0.373	0.404	0.823	1.139	0.988
	RGC	1.064	0.777	0.875	0.152	0.685	0.294	0.409	0.501	0.579	1.016
	UMRF	0.120	0.576	0.176	0.133	0.701	0.249	0.327	0.425	0.498	0.747
BDE	GC	30.56	24.04	15.98	4.39	15.98	18.69	11.28	9.72	7.98	7.84
	ICM	28.90	10.03	8.25	3.39	16.25	13.27	10.39	9.80	17.87	9.69
	MRMRF	34.52	10.96	15.94	5.38	10.85	16.09	8.18	10.03	18.05	10.29
	SMRF	26.48	18.14	16.06	4.73	18.74	22.90	7.77	10.89	16.61	10.75
	RGC	21.05	12.94	16.94	0.66	15.28	18.68	6.40	6.16	12.87	11.64
	UMRF	3.57	7.52	0.62	1.19	15.14	14.40	3.26	3.20	2.96	8.52

and the snow land in (*d*), the snake and the desert in (*e*), the hair of the man on the left and the sea in (*f*), the leaves and the sky in (*m*). These reasons make the segmentation challenging.

The test results of the UMRF model are shown in Fig. 10. One can see that the UMRF model performs well and the results are close to the ground truth. This is due to the fact that the UMRF model is capable of capturing various types of important features. Furthermore, the quantitative indexes of these experiments are shown in Table 2.

The general selection criteria for α and β of the UMRF model are discussed in Section 4.3. In practice, α often takes value around 1.5. Moreover, it could take a smaller value to emphasise the pixel feature; on the contrary, it could take a larger value to emphasize the regional feature. Parameter β usually takes value around 1. One can stress the $P(X)$ by increasing beta value, conversely, one can

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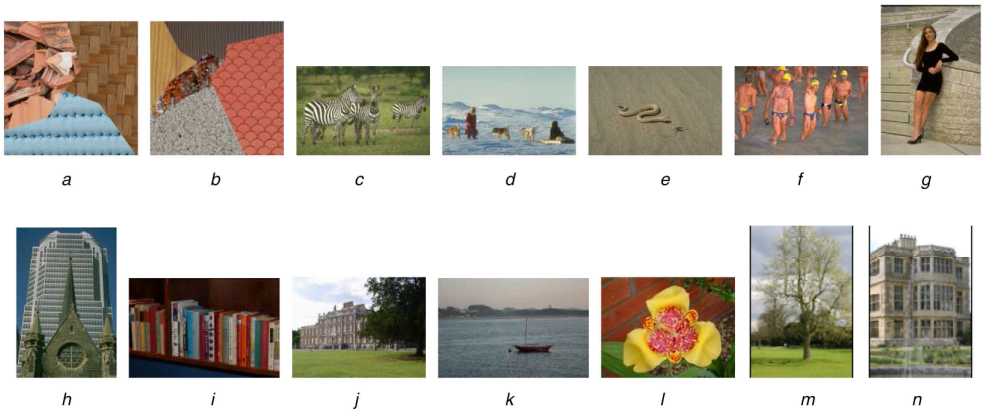


Fig. 9 *Tested images from different image datasets*
(a), ***(b)*** Images from the Prague texture segmentation datagenerator, ***(c)–(h)*** Images from the Berkeley segmentation dataset, ***(i)–(n)*** Images from the Microsoft dataset (MSRC2)

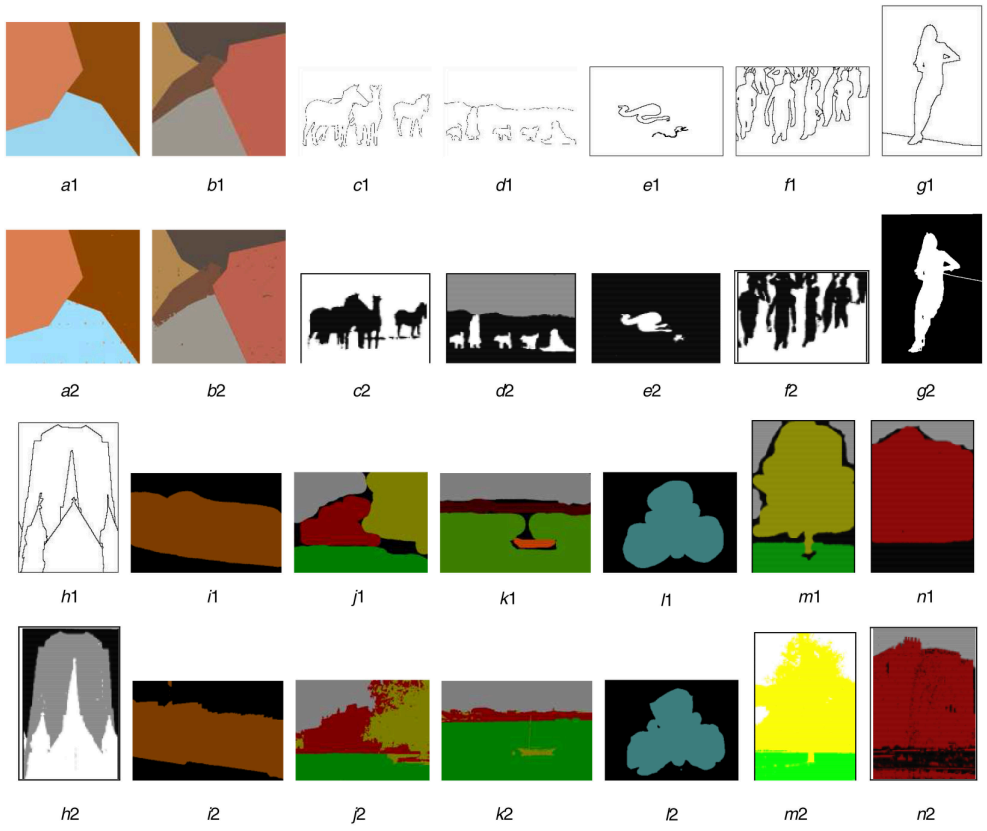


Fig. 10 *Segmentation results of UMRF and the ground truth*
(a1)–(n1) Ground truth, ***(a2)–(n2)*** Results obtained from UMRF

Table 2 The values of quantitative indexes for experiments of Fig. 10

	a	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*i*	*j*	*k*	*l*	*m*	*n*
PRI	0.985	0.966	0.885	0.974	0.921	0.871	0.943	0.928	0.903	0.889	0.931	0.946	0.818	0.787
VoI	0.006	0.036	0.104	0.036	0.052	0.127	0.051	0.101	0.094	0.192	0.087	0.052	0.184	0.221
GCE	0.259	0.864	0.594	0.262	0.377	0.712	0.323	0.646	0.556	1.238	0.742	0.314	1.208	1.218
BDE	3.449	5.769	14.41	12.91	23.83	3.654	7.070	3.447	5.038	11.02	2.844	2.372	10.35	15.54

stress the $P(\boldsymbol{Y}|\boldsymbol{X})$ by decreasing beta value. The parameter values of experiments for Figs. 7 and 10 are all shown in Table 3.

7 Conclusion

A UMRF model for image segmentation is proposed in this paper, which unifies advantages of the pixel- and the region-based MRF model. There are two mainly contributions of the UMRF model. First, it employs various types of features, such as the spectral value for the pixel feature, the region size and the region context for the regional feature. Second, this model can incorporate these various types of information by decomposing the likelihood function into the product of the pixel likelihood function and the region likelihood function. Furthermore, one can flexibly use different parameters to adjust the contribution of these two likelihood functions. Favourable results can be achieved when

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Table 3a *Continued*

Fig. 7	a5	b5	c5	d5	e5	f5	g5	h5	i5	j5
α	1.1	2	0.25	3	1.5	1.5	0.6	3.6	0.65	5
β	0.15	0.25	0.25	5	0.5	0.5	0.5	4	0.5	5

Table 3b Parameter settings for experiments of the UMRF model in Figs. 7 and 10

Fig. 10	a2	b2	c2	d2	e2	f2	g2	h2	i2	j2	k2	l2	m2	n2
α	3.75	0.25	0.75	0.1	0.2	2	1	0.4	1.75	1.2	0.3	2.5	1	0.5
β	1	0.4	1.5	2	2	1	2.5	0.5	1	1	0.3	5	0.5	0.2

appropriate parameters are selected based on the images. The experimental results obtained from various datasets demonstrate that the UMRF model can show a good performance and improve the segmentation accuracy. Furthermore, some parameters of the UMRF model, such as α and β, need to be set empirically. If these parameters can be estimated automatically, segmentation results with higher accuracy may be achieved.

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